



Project. 1

Training: Network Infrastructure

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UNDER SUPERVISION OF

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1. Introduction

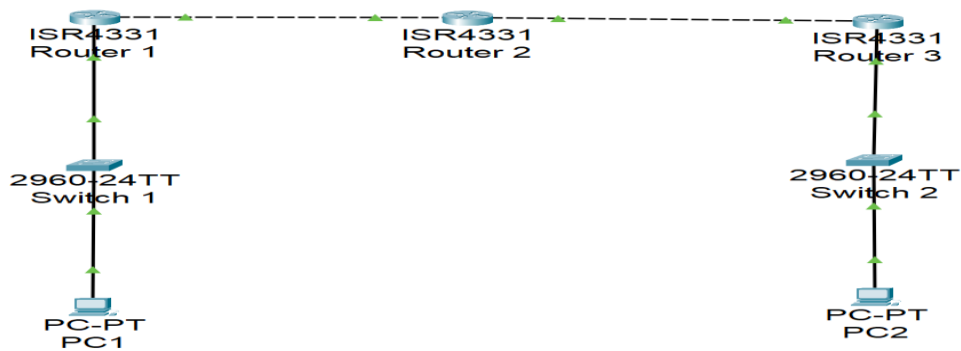
The purpose of this project is to design and implement a computer network in Cisco Packet Tracer using three routers, two switches, and two PCs. The project includes IP addressing, DHCP configuration, static addressing, OSPF routing, remote access (Telnet & SSH), and device security.

2. Network Topology

The network topology consists of:

- Routers: R1, R2, R3
- Switches: S1, S2
- PCs: PC1, PC2

The topology diagram shows the connections between the devices.



3. IP Addressing

Device/Interface	IP Address	Subnet Mask	Notes
R1 G0/0/0	192.168.1.1	255.255.255.0	LAN (DHCP)
R1 G0/0/1	20.0.0.1	255.255.255.252	Link to R2
R2 G0/0/0	20.0.0.2	255.255.255.252	Link to R1
R2 G0/0/1	30.0.0.1	255.255.255.252	Link to R3
R3 G0/0/1	30.0.0.2	255.255.255.252	Link to R2
R3 G0/0/0	10.0.0.1	255.255.255.0	LAN (Static)
PC1	DHCP	-	Gets IP from R1

PC2	10.0.0.2	255.255.255.0	Static, GW=10.0.0.1
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4. Configurations

The following configurations were applied on the devices:

R1

```
hostname R1
int g0/0/0
ip address 192.168.1.1 255.255.255.0
no shut
int g0/0/1
ip address 20.0.0.1 255.255.255.252
no shut
ip dhcp excluded-address 192.168.1.1 192.168.1.10
ip dhcp pool LAN1
network 192.168.1.0 255.255.255.0
default-router 192.168.1.1
dns-server 8.8.8.8
router ospf 1
network 192.168.1.0 0.0.0.255 area 0
network 20.0.0.0 0.0.0.3 area 0
username admin password 123
ip domain-name test.com
crypto key generate rsa modulus 1024
line vty 0 4
login local
transport input ssh
```

R2

```
hostname R2
int g0/0/0
ip address 20.0.0.2 255.255.255.252
no shut
int g0/0/1
ip address 30.0.0.1 255.255.255.252
no shut
router ospf 1
network 20.0.0.0 0.0.0.3 area 0
```

```
network 30.0.0.0 0.0.0.3 area 0
```

R3

```
hostname R3
int g0/0/0
ip address 10.0.0.1 255.255.255.0
no shut
int g0/0/1
ip address 30.0.0.2 255.255.255.252
no shut
router ospf 1
network 10.0.0.0 0.0.0.255 area 0
network 30.0.0.0 0.0.0.3 area 0
line vty 0 4
password cisco
login
transport input telnet
```

S1

```
hostname S1
enable secret class
line con 0
password cisco
login
```

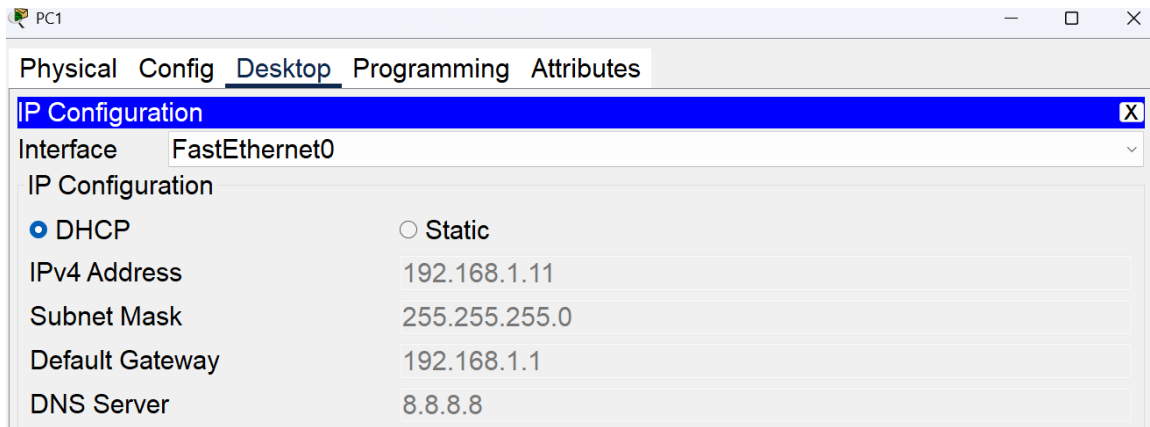
S2

```
hostname S2
enable secret class
line con 0
password cisco
login
```

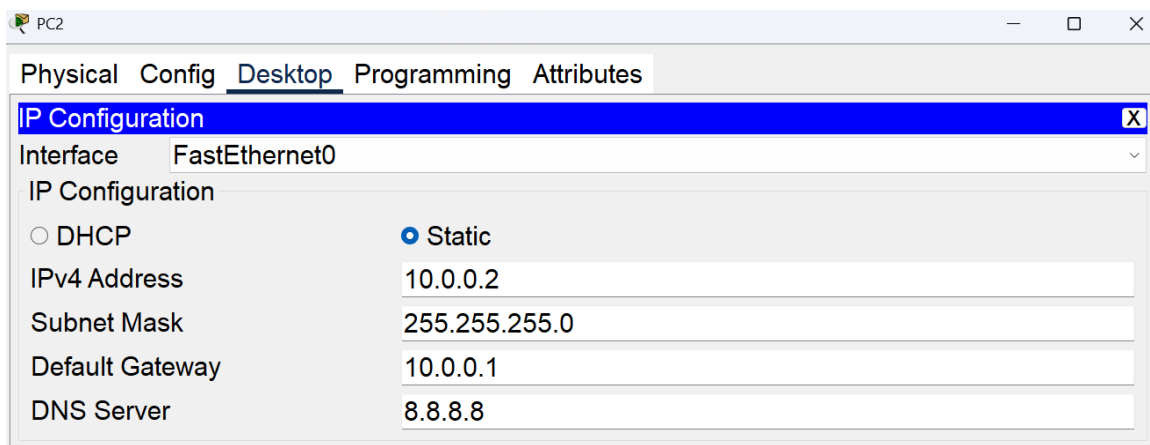
5. Verification

The following tests were conducted to verify the network configuration:

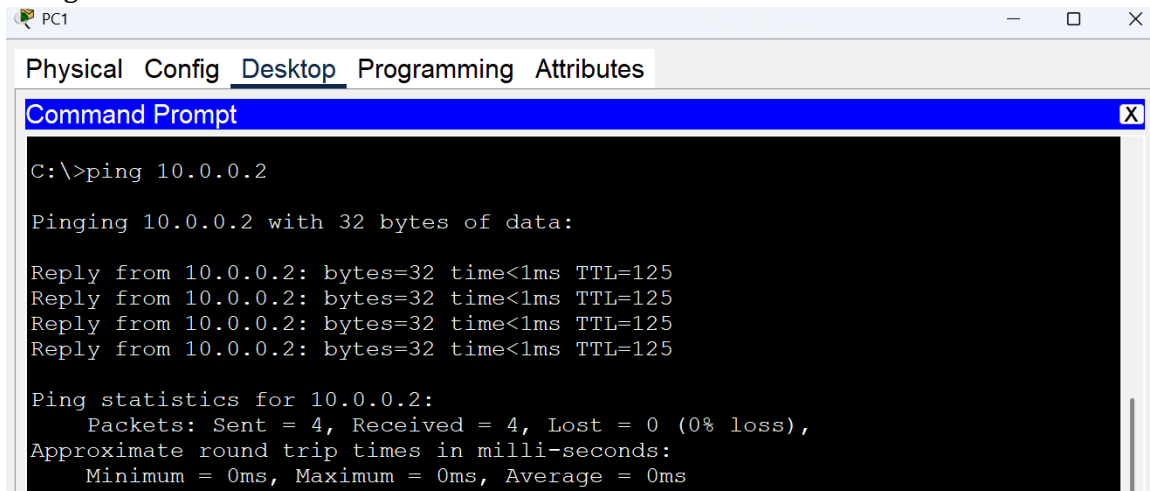
- PC1 successfully obtained an IP address from DHCP on R1.

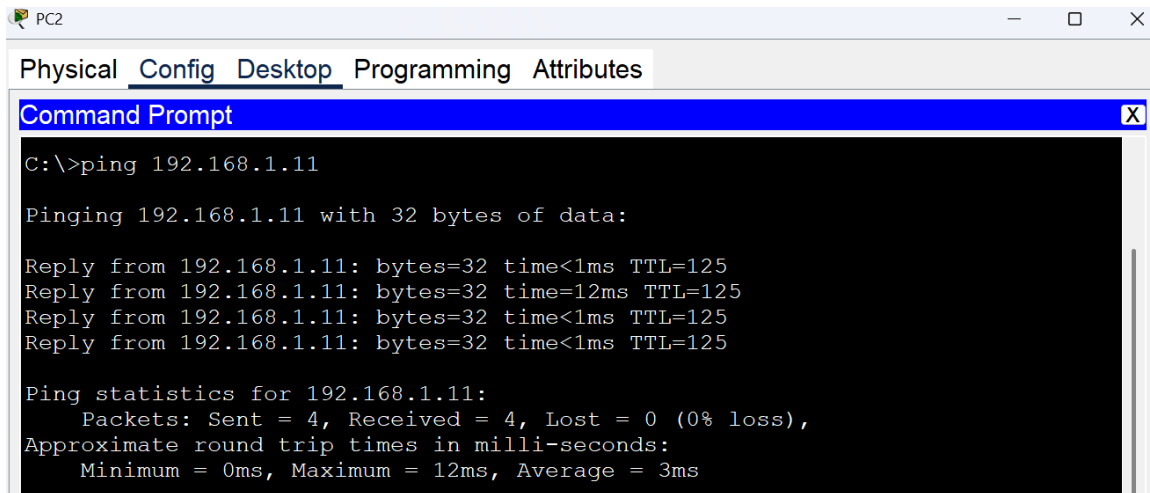


- PC2 was configured with a static IP address (10.0.0.2).



- Ping test between PC1 and PC2 was successful.





PC2

Physical Config Desktop Programming Attributes

Command Prompt

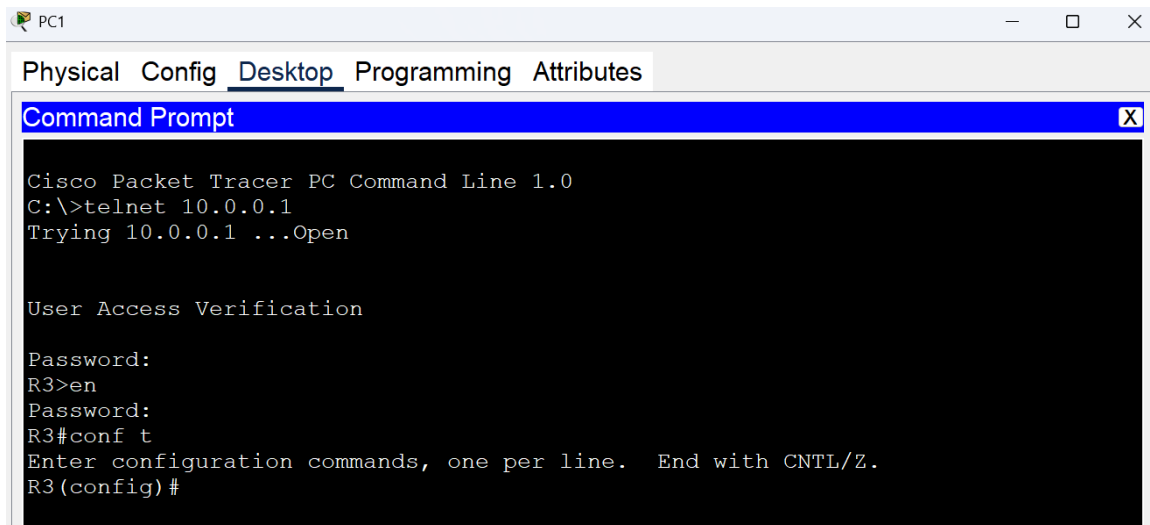
```
C:\>ping 192.168.1.11

Pinging 192.168.1.11 with 32 bytes of data:

Reply from 192.168.1.11: bytes=32 time<1ms TTL=125
Reply from 192.168.1.11: bytes=32 time=12ms TTL=125
Reply from 192.168.1.11: bytes=32 time<1ms TTL=125
Reply from 192.168.1.11: bytes=32 time<1ms TTL=125

Ping statistics for 192.168.1.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 12ms, Average = 3ms
```

- Telnet from PC1 to R3 was successful.



PC1

Physical Config Desktop Programming Attributes

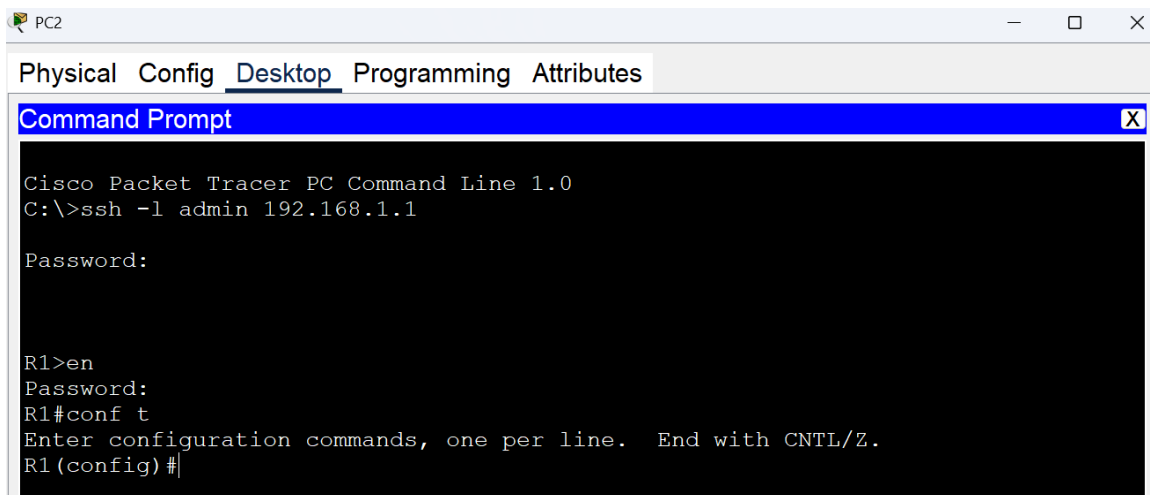
Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>telnet 10.0.0.1
Trying 10.0.0.1 ...Open

User Access Verification

Password:
R3>en
Password:
R3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#
```

- SSH from PC2 to R1 was successful.



PC2

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ssh -l admin 192.168.1.1

Password:

R1>en
Password:
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#
```

6. Conclusion

This project provided hands-on experience in configuring DHCP, static addressing, OSPF routing, and secure remote access on Cisco routers and switches. It reinforced practical networking skills and simulated real-world network management tasks.

7. Project link

https://drive.google.com/file/d/1aAa8Bv5H5gDr2xicjTBtfDGHl4yXlBiq/view?usp=drive_link